THE TITLE

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Abstract

THE ABSTRACT.

Keywords: KEYWORD, KEYWORD, KEYWORD

THE TITLE

Binge drinking is bad, it leads to a lot of bad things, and some epi stats, which means we're important.

Binge drinking interventions often try to target...something.

But those somethings are often poorly operationalized or not at all, either through...idk...probably self-report. Studies on self-report show (what, i imagine they're pretty bad, but maybe they're actually surprisingly good). Wearable transdermal alcohol sensors have emerged as an alternative to self-report, offering a passive continuous measure of alcohol consumption (Fairbairn et al., 2025; Gunn et al., 2023). It surpasses previous measures of intoxication (eg breathalyzers) by reducing the burden traditionally involved in obtaining a measurement to (blakn) what were the findings of those feasability studies? But although we have all this data, we dont know what to do with it. Raw transdermal alcohol concentration (TAC) signal can be difficult to interpret and translate into meaningful drinking parameters (Gunn et al., 2023; Kianersi et al., 2023).

To bridge the gap between theory and methods—because this phrasing looks good for pamt— , we propose a formal model for binge-drinking, which operationalizes binged drinking by...—some number— parameters characterizing...—the things we characterize—.

Drawing from evidence in...some binge-drinking applied literature...we conceptualize binge-drinking as...the empirical phenomena that make up binge-drinking... For example, TAC-derived features such as rise rate, peak concentration, and rise duration have been shown to predict alcohol-induced blackouts in college drinkers (Richards et al., 2024), and TAC signal processing models have been developed and validated against self-reported and breath alcohol concentration data in similar populations (Kianersi et al., 2023). Our model uses ideas from...—insert literature, ie michaelis menten or that one dynamical systems model of loneliness— ...to characterize—the big picture formulation ie between drinking time, within episode duration, within episode drinking patterns—as a dynamical system. Compared to previous models, ours provide interpretable parameters with mechanistic meanings. Furthermore, we are able to distinguish between the end of a drinking episode and a return to baseline TAC. We either if its not estimatable, some estimation routine speed up or contribution if its actually fine, an in depth demonstration and evaluation of the model utility

simulation demonstrate the accuracy of our method/software.

We illustrate how these parameters could be used to characterize the effects of risk factors on binge-drinking in college students.

the remainder of this paper is organized as follows: —big long run-on sentence—

the model

Simulation Method

Describe the design of the simulation study: the data-generating model, the conditions manipulated, and the number of replications per condition.

Data-Generating Model

Specify the model used to generate the simulated data, including all fixed parameter values.

Design and Conditions

Describe the factors crossed in the simulation design (e.g., sample size, number of measurement occasions, effect size) and their levels.

Estimation and Evaluation Criteria

Describe the estimator(s) fit to each replication, the software used, and the criteria by which performance is evaluated (e.g., relative bias, RMSE, confidence interval coverage, convergence rates).

Simulation Results

Report the results of the simulation study, using APA-style statistical reporting throughout. Convergence rates and performance measures appear in Table 1, and Figure 1 shows how these measures vary with sample size and effect size.

Empirical Methods

Participants

Describe your sample: size, demographics, recruitment method, and any inclusion/exclusion criteria.

Materials

Describe instruments, apparatus, or materials used.

Procedure

Describe the step-by-step procedure of the study.

Data Analysis

Describe your analytic strategy (e.g., statistical tests, software used).

Empirical Results

Report your findings. Use APA-style statistical reporting, e.g., a significant effect was found, $t(28) = 3.45$, $p < .001$, $d = 0.65$. Regression estimates appear in Table 2, and Figure 2 shows the pattern of results.

Discussion

What were your main aims? What were key findings regarding each aim?

Compare and contrast findings observed to past literature.

Limitations and Future Directions

At least 2 limitations. How do your findings contribute to existing literature?

How should we move forward in research or practice given your contribution to the literature?

Conclusion

Summarize the key takeaways of your paper.

References

Fairbairn, C. E., Han, J., Caumiant, E. P., Benjamin, A. S., & Bosch, N. (2025). A wearable alcohol biosensor: Exploring the accuracy of transdermal drinking detection. *Drug and Alcohol Dependence*, *266*, 112519. https://doi.org/10.1016/j.drugalcdep.2024.112519

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Table 1

Simulation Performance Measures

N	True β_1	Bias	RMSE	Coverage	Power	Convergence
50	0.00	-0.001	0.143	0.964	0.036	1.000
50	0.20	-0.001	0.135	0.952	0.697	1.000
50	0.50	-0.001	0.142	0.951	1.000	1.000
100	0.00	0.003	0.100	0.946	0.045	1.000
100	0.20	0.001	0.104	0.950	0.799	1.000
100	0.50	0.003	0.096	0.948	1.000	1.000
250	0.00	0.004	0.063	0.952	0.051	1.000
250	0.20	0.004	0.061	0.953	1.000	1.000
250	0.50	-0.013	0.062	0.945	1.000	1.000

Note. Bias, RMSE, and 95% CI coverage/power are computed over converged replications only.

Convergence is the proportion of replications that converged.

Table 2

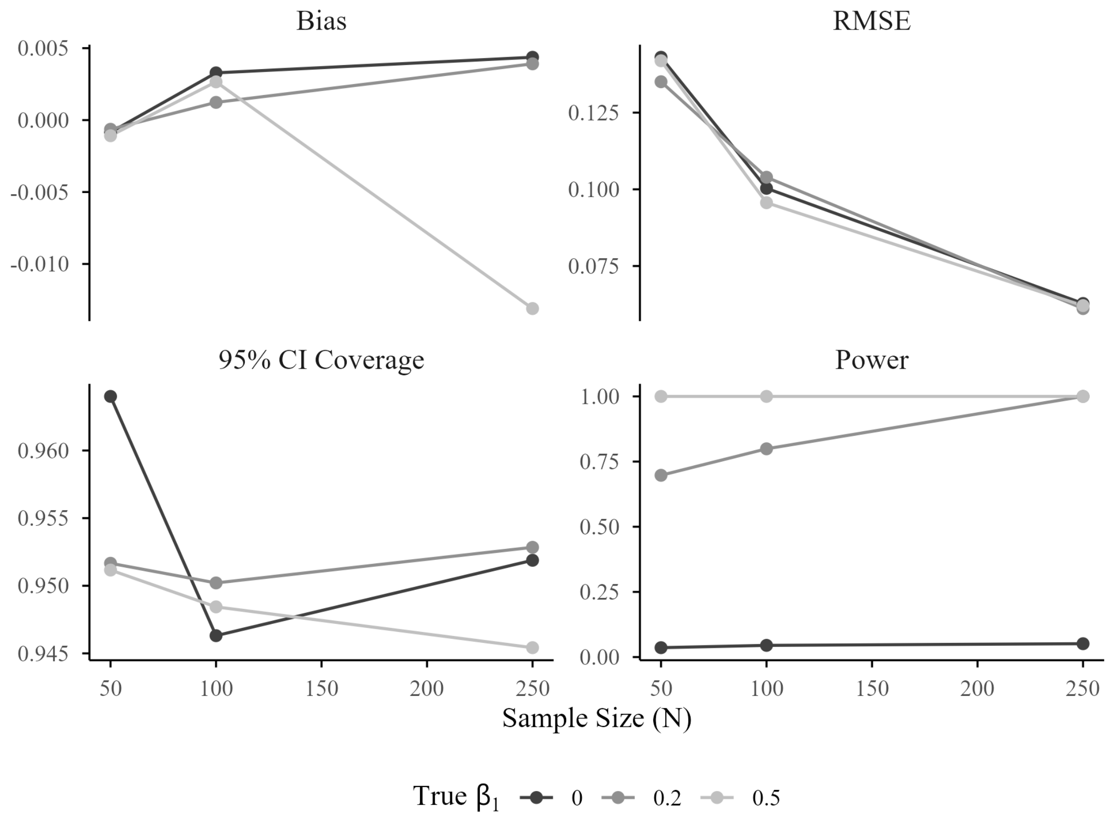
Empirical Analysis: Regression Estimates

Term	*B*	*SE*	*p*	95% CI
Intercept	7.721	3.162	0.015	[1.491, 13.951]
X_1	0.369	0.060	0.000	[0.250, 0.487]
Treatment	2.133	1.122	0.059	[-0.077, 4.344]

Note. N = 233, R^2 = 0.154.

Figure 1

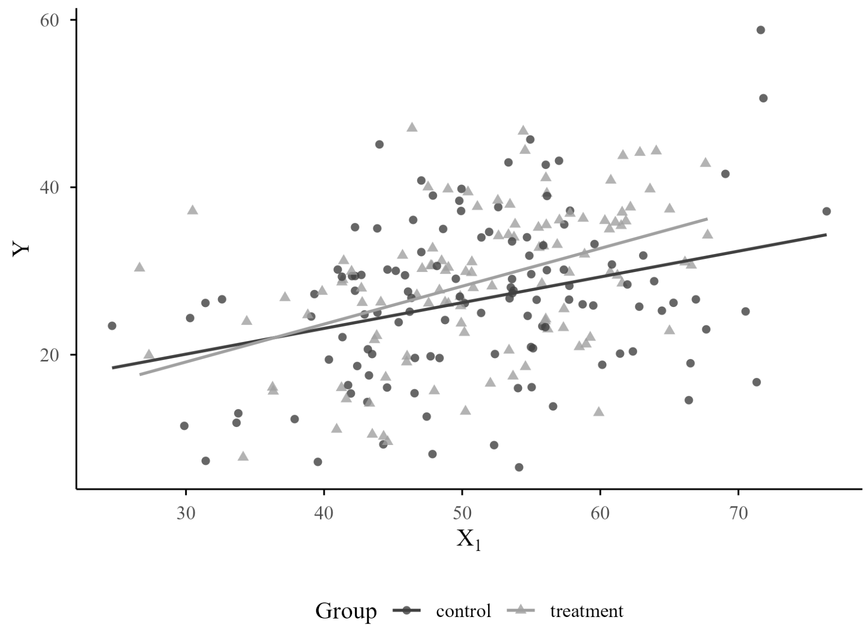
Simulation Performance Measures by Sample Size and Effect Size



Note. Bias, RMSE, and 95% CI coverage/power are computed over converged replications only.

Figure 2

Outcome by Predictor and Group



Note. Points are individual observations; lines are group-wise fitted values.